

PRACTICAL - 2

AIM:RSA Encryption and Decryption

Implement the RSA algorithm for public-key encryption and decryption, and explore its properties and security considerations.

CLI:

CODE:

```
from math import gcd

# Function to find modular inverse
def mod_inverse(e, phi):
    for d in range(2, phi):
        if (d * e) % phi == 1:
            return d
    return None

# Input two prime numbers
p = int(input("Enter first prime number (p): "))
q = int(input("Enter second prime number (q): "))

# Calculate n and phi
n = p * q
phi = (p - 1) * (q - 1)

# Choose e
e = int(input("Enter value of e: "))

if gcd(e, phi) != 1:
    print("e must be co-prime with phi(n).")
    exit()

# Calculate d
d = mod_inverse(e, phi)

print("\nPublic Key :", (e, n))
print("Private Key:", (d, n))

# Encrypt
msg = int(input("\nEnter message (integer less than n): "))
cipher = pow(msg, e, n)
print("Encrypted Message:", cipher)

# Decrypt
plain = pow(cipher, d, n)
print("Decrypted Message:", plain)
```

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OUTPUT:

```
IDLE Shell 3.14.6
File Edit Shell Debug Options Window Help
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>> ===== RESTART: C:\Users\info\OneDrive\Desktop\CyberSecurity_Prac2.py =====
Enter first prime number (p): 11
Enter second prime number (q): 13
n = 143
phi = 120
Enter value of e: 7

Public Key : (7, 143)
Private Key: (103, 143)

Enter message (integer less than n): 9
Encrypted Message: 48
Decrypted Message: 9

>>>
```

```
CA\WINDOWS\py.exe
Enter first prime number (p): 11
Enter second prime number (q): 13
n = 143
phi = 120
Enter value of e: 7

Public Key : (7, 143)
Private Key: (103, 143)

Enter message (integer less than n): _
```

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GUI:

CODE:

```
import tkinter as tk
from tkinter import messagebox
from math import gcd

# Function to find modular inverse
def mod_inverse(e, phi):
    for d in range(2, phi):
        if (d * e) % phi == 1:
            return d
    return None
```

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```
def generate_keys():
    global n, e, d

    try:
        p = int(entry_p.get())
        q = int(entry_q.get())
        e = int(entry_e.get())

        n = p * q
        phi = (p - 1) * (q - 1)

        if gcd(e, phi) != 1:
            messagebox.showerror("Error", "e must be co-prime with phi(n)")
            return

        d = mod_inverse(e, phi)

        lbl_public.config(text=f"Public Key: ({e}, {n})")
        lbl_private.config(text=f"Private Key: ({d}, {n})")

    except:
        messagebox.showerror("Error", "Enter valid numbers.")

def encrypt():
    try:
        msg = int(entry_msg.get())

        if msg >= n:
            messagebox.showerror("Error", f"Message must be less than {n}")
            return

        cipher = pow(msg, e, n)
        entry_cipher.delete(0, tk.END)
        entry_cipher.insert(0, str(cipher))

    except:
        messagebox.showerror("Error", "Generate keys first.")

def decrypt():
    try:
        cipher = int(entry_cipher.get())
        plain = pow(cipher, d, n)

        entry_plain.delete(0, tk.END)
        entry_plain.insert(0, str(plain))
```

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```
except:  
    messagebox.showerror("Error", "Invalid Cipher Text.")
```

```
#GUI
```

```
root = tk.Tk()  
root.title("RSA Encryption & Decryption")  
root.geometry("450x400")
```

```
tk.Label(root, text="RSA Encryption & Decryption",  
         font=("Arial", 16, "bold")).pack(pady=10)
```

```
frame = tk.Frame(root)  
frame.pack()
```

```
tk.Label(frame, text="Prime p").grid(row=0, column=0, padx=5, pady=5)  
entry_p = tk.Entry(frame)  
entry_p.grid(row=0, column=1)
```

```
tk.Label(frame, text="Prime q").grid(row=1, column=0, padx=5, pady=5)  
entry_q = tk.Entry(frame)  
entry_q.grid(row=1, column=1)
```

```
tk.Label(frame, text="Public Key e").grid(row=2, column=0, padx=5, pady=5)  
entry_e = tk.Entry(frame)  
entry_e.grid(row=2, column=1)
```

```
tk.Button(root, text="Generate Keys",  
          command=generate_keys,  
          bg="lightblue").pack(pady=10)
```

```
lbl_public = tk.Label(root, text="Public Key:")  
lbl_public.pack()
```

```
lbl_private = tk.Label(root, text="Private Key:")  
lbl_private.pack()
```

```
tk.Label(root, text="Message").pack()  
entry_msg = tk.Entry(root)  
entry_msg.pack()
```

```
tk.Button(root, text="Encrypt",  
          command=encrypt,  
          bg="lightgreen").pack(pady=5)
```

```
tk.Label(root, text="Cipher Text").pack()
```

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```
entry_cipher = tk.Entry(root)
entry_cipher.pack()

tk.Button(root, text="Decrypt",
          command=decrypt,
          bg="orange").pack(pady=5)

tk.Label(root, text="Decrypted Message").pack()
entry_plain = tk.Entry(root)
entry_plain.pack()

root.mainloop()
```

OUTPUT:

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RSA Encryption & Decryption

RSA Encryption & Decryption

Prime p

Prime q

Public Key e

Public Key: (7, 483)
Private Key: (63, 483)

Message

Cipher Text

Decrypted Message

Windows taskbar: Search, Taskbar icons, System tray: ENG IN, 08:21, 20-07-2026